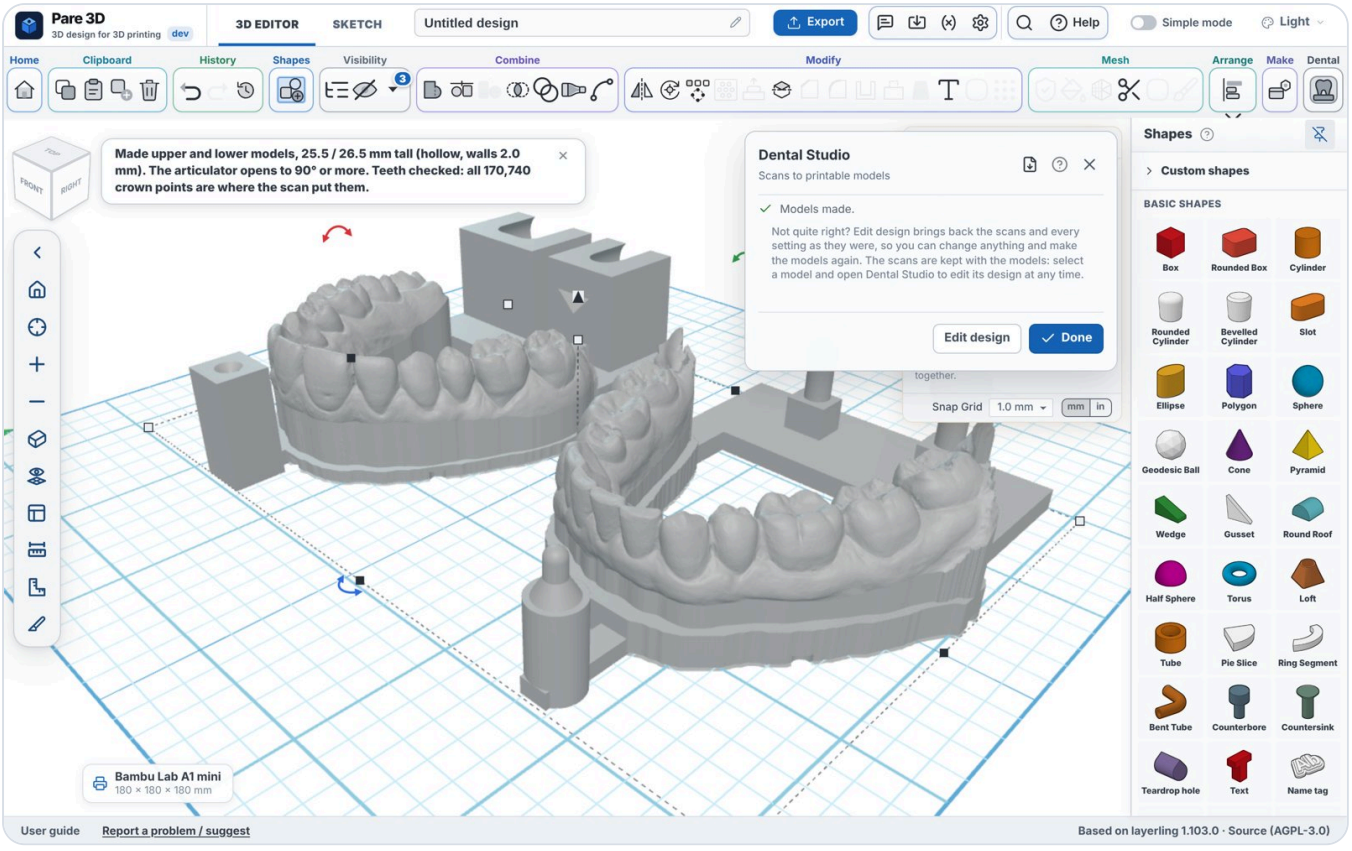
Pare 3D · Dental Studio · Build 80 · October 2026

# The Model Builder Guide

From intraoral scans to printed study, ortho, aligner, crown-and-bridge and articulated models - six steps, every setting explained for clinicians.




An upper and lower pair made from three intraoral scan files, with the articulator: the condyles behind the molars and the incisal pin in front, in its cup, laid out ready to print.

Screens shown are from Pare 3D build 80 on a 1440 × 900 window. Scans used: a full upper arch (330,577 triangles), a full lower arch and a buccal bite scan from the author's own mouth.

# Contents

1 What Dental Studio does
2 Before you start: scans and files
3 Opening Dental Studio
4 Step 1 - Add the scans
5 The bite: one bite scan, two, or none
6 Step 2 - Prepare the scans: trim and fill, then check the teeth
7 Step 3 - Choose the model
8 Step 4 - The base and the model's height
9 Step 5 - The articulator
10 Step 6 - Name the model
11 Making the models, and editing the design
12 Editing a scan by hand: Sculpt
13 Printing
14 Messages and what to do
15 Quick reference
16 Accuracy, and checking a printed articulator
17 References

**The short way**

Home page › **Dental models from scans** › drag the upper, lower and bite files onto the window › pick what the model is for › tick **Print as an articulator** if you want one › **Make models** › **Send to** your slicer. Everything else in this guide is for when you want to change what Pare 3D chose.

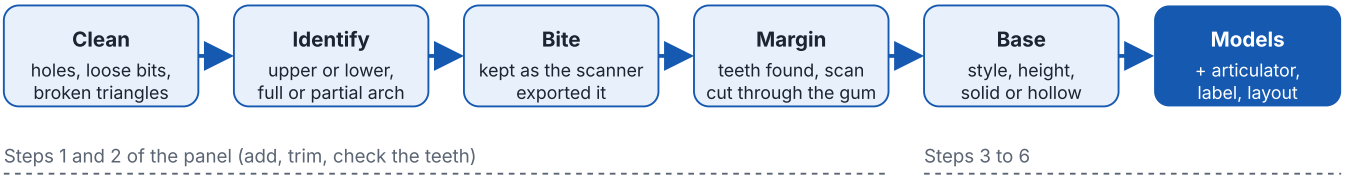
**New in builds 79 and 80**

The articulator locates like a lab articulator - the pin's tip in a cup - or with a post either side, clear of the front teeth; the pin is part of the model and never touches a tooth (chapter 9). One or two bite scans (chapter 5). Edit design after making, and Stop while making (chapter 11). Scans and models in grey. (The digital wax-up shown in build 79 is being developed further as a module of its own, in test builds.)

# 1 What Dental Studio does

Dental Studio turns raw intraoral scans - open surfaces that end where the scanner stopped - into closed, printable models with a base, and, for a pair, matching bases in the bite and an optional articulator.

It does the work a dental lab technician or a model-builder program (Medit Model Builder, 3Shape Model Builder) does, in one panel and in this order:




The first four happen by themselves as soon as the scans are in; you check them and change what you want.

## What is never changed

- **The tooth surface.** The scanned teeth and the gum beside them are kept exactly as scanned. They are only turned and moved as a whole; no point of the kept surface moves relative to another. Trimming and margin cuts never cross a tooth.
- **The bite.** Arches that arrive in occlusion stay exactly where the scanner put them: the jaw relationship is the one your scanning software recorded. Only if you ask is each arch fitted to the bite scan, by one rigid transform each - and then that fit is locked in (chapter 5).
- **Checked, every time.** Every crown point of each scan is checked against the finished model before it is handed over, and the notice says so (chapter 16).

## What you get

- One model per arch, closed and printable, named after the scan (*<scan name> upper model*, *<scan name> lower model*), with a base of the kind you chose.
- For a pair: matching bases that stand in the bite, laid out side by side on the build plate or stacked in occlusion, and - if you ask - an articulator built into the bases.
- A notice saying what was made (heights, hollow or solid, wall thickness) and any warnings.

**Where it runs**

Everything happens on your own computer, in the Pare 3D app or browser tab; scans are never uploaded. An articulated pair from full-arch scans took about 50 s on a small 2-core test machine; a recent desktop is faster. The window stays responsive while it works.

# 2 Before you start: scans and files

Dental Studio needs the arch scans as surface meshes, in millimetres, exported from the same case so they share the scanner's coordinates.

## File types

| Format | Use | Notes |
|---|---|---|
| STL | Most common export from all scanners | Geometry only. Pare 3D reads binary and text STL. |
| PLY | Medit, 3Shape and others export colour this way | Read as geometry; scan colour is not used for the model. |
| OBJ | Some scanners and labs | Geometry is used; textures are ignored. |
| 3MF | Exported from other programs | Each mesh in it comes in as a part. |

## What to export from the scanner

1. **Upper arch** - the maxillary scan.
2. **Lower arch** - the mandibular scan.
3. **Bite scan** (optional but recommended for a pair) - the buccal bite registration: the scan of the teeth in occlusion from the buccal side. Most scanners export it as a third file, or include it in the case export.
4. **Second bite scan** (optional, recommended) - many clinicians record the bite on each side. Export both; a bilateral record aligns the whole arch more accurately than one side (chapter 5).

> **Export them all from the same case, without moving them**
> Scanners export the arches and the bite in one shared coordinate system, usually already in occlusion. Dental Studio starts from those positions. A scan that was moved, mirrored or re-oriented in another program before export may not line up; Pare 3D then says so (*The two scans are not where the scanner put them...*) and makes each arch on its own.

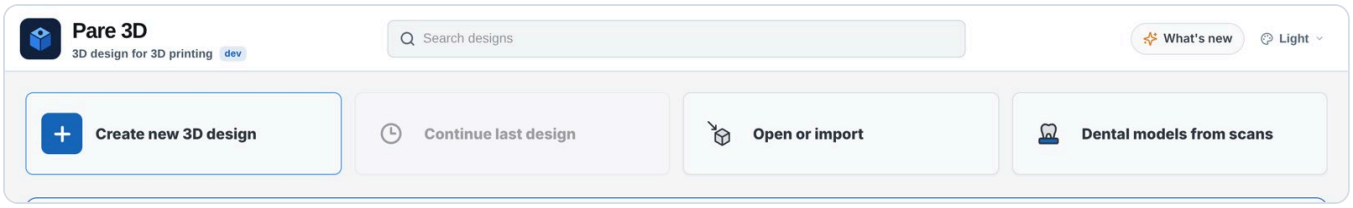
## Scan quality that matters

- **Captured depth.** Scan far enough into the vestibule and lingual sulcus for the base you want. The panel reports how many millimetres of scan lie below the occlusal plane for each arch; an ortho base needs the most.
- **Holes and loose islands.** Small holes are filled and loose fragments are removed automatically. Large gaps on the teeth themselves cannot be invented - rescan those areas.
- **Partial arches** (a quadrant or sextant) are recognised as partial and get a block base; the tie bar option is for full arches only (chapter 4, *Full arch or quadrant*).
- **Units.** Files must be in millimetres. A scan that comes in a few millimetres or several metres across was exported in another unit.

> **Patient files**
> Name files with a patient code rather than the patient's name if the designs or exports will be shared. The Label option (chapter 10) engraves a code on the model.

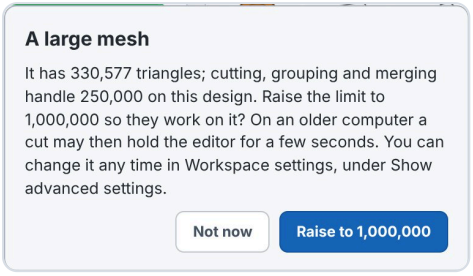
# 3 Opening Dental Studio

There are three ways in; all end in the same panel.




The home page. **Dental models from scans** opens a new design with Dental Studio already open; **What's new** lists what each build brought.

1. **From the home page** - click **Dental models from scans**. A new, empty design opens with Dental Studio ready (fastest for a new case).
2. **By dropping the files** - drag the scan files from your desktop or a folder onto the editor window. A dashed frame shows while you hold them over it; they come in when you let go. Pare 3D recognises intraoral scans and offers **Dental Studio** in a notice at the bottom of the window.
3. **From the toolbar** - the tooth button in the **Dental** group, or search `Ctrl` + `K` for *dental*, *articulator* or *model builder*. With scans selected, those are used; with none selected, every scan in the design is offered.

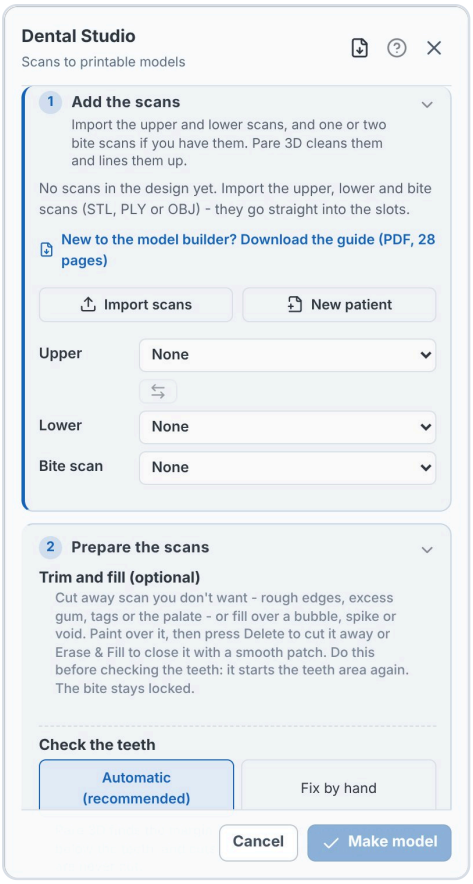



Offered once a session when a scan has more triangles than cutting and grouping handle.

## A large mesh: raise the limit once

Cutting, grouping and merging an imported mesh are worked out in the browser, up to a limit of 250,000 triangles by default. A full-arch scan has more (330,577 for the upper used here). When one comes in, a card offers to raise the limit for the design to room for twice the mesh - at least 1,000,000. Press **Raise to 1,000,000**: Dental Studio itself does not need it, but grouping a model with a label or cutting it with a hole does. On an older computer a cut can then hold the editor for a few seconds.

The card asks once a session. The setting stays in **Workspace settings** under **Show advanced settings** (*Cut imported meshes up to*), to change at any time.



Dental Studio before any scan is in.

## The panel: six step cards

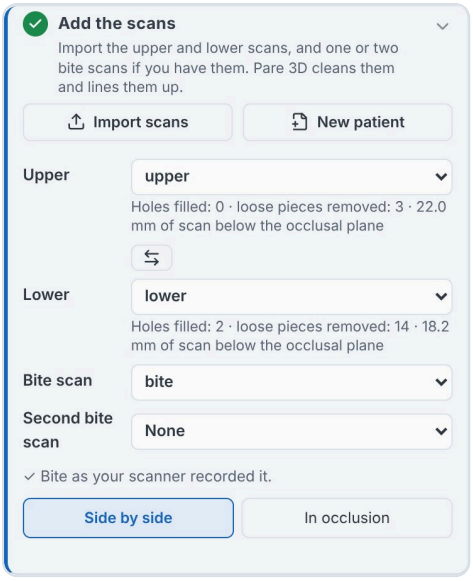
Each step sits on a card of its own, top to bottom. A step's number turns into a green tick once it is complete and you have worked in it. Once the scans are in, the step to do next is outlined in green until you have worked in it. Click a step's title to fold it to one line that sums up its settings, and again to open it; the card you last worked in is marked with a blue edge. The line above the buttons says what is still needed - for example *All set - press Make models*.

1. **Add the scans** - which scan is upper, lower and bite (one or two); side by side or in occlusion.
2. **Prepare the scans** - trim or fill what you do not want, then check the teeth.
3. **Choose the model** - what it is for, and solid or hollow.
4. **Base** - its shape, details and the model's height.
5. **Articulator** - only shown for a pair.
6. **Name the model** - a label, or no patient details.

With a single arch the articulator card is left out, so there are five. The panel can be dragged by its title bar; the download button beside **?** saves this guide as a PDF (also linked in step 1 before any scan is in); **?** opens this part of the user guide; **×** or **Cancel** closes it without changing anything.

# 4 Step 1 - Add the scans

Bring the files in, check that each landed in the right slot, and read the cleaning report under each arch.



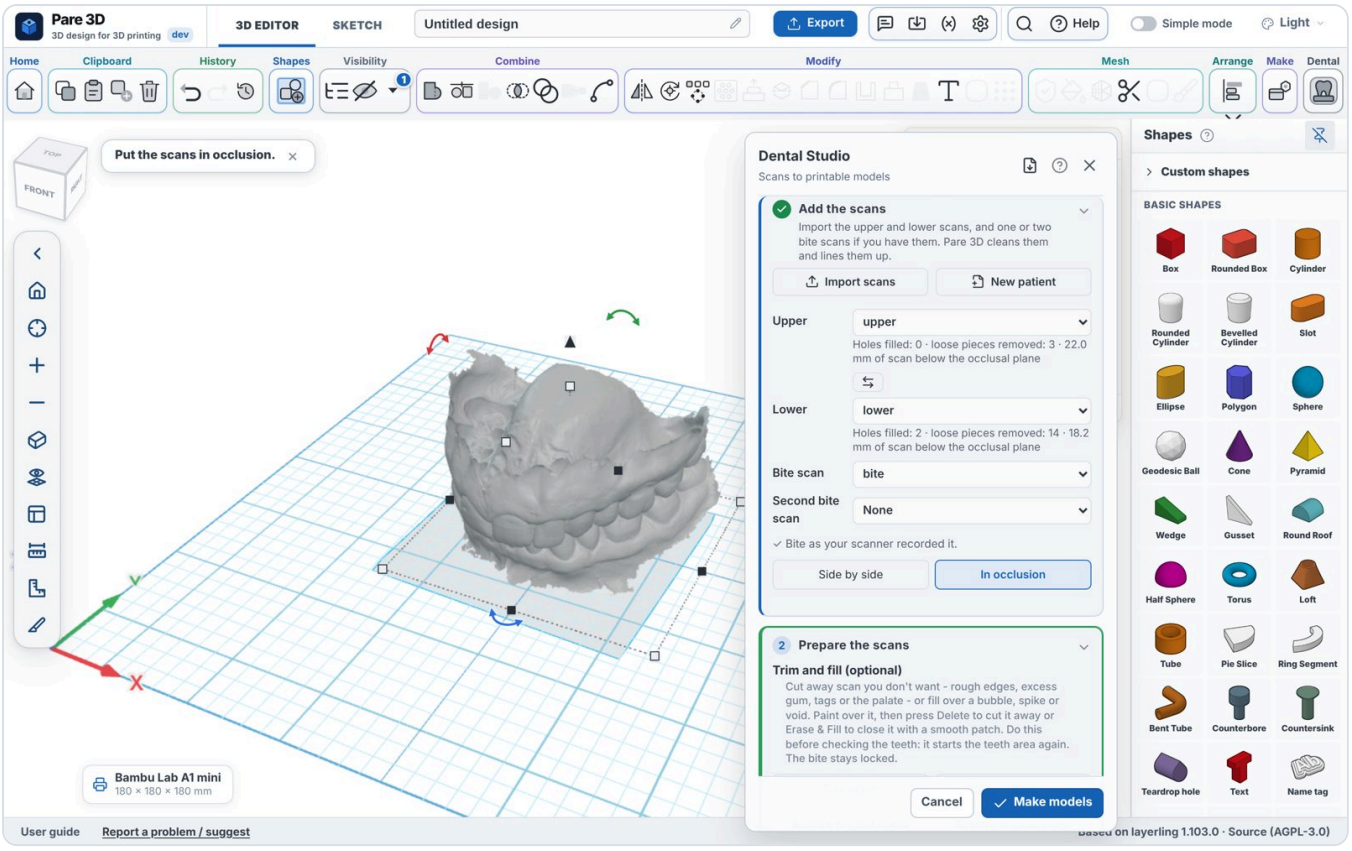

Upper, lower and bite filled; the scanner's bite is kept, with a tick.

1. Drag the upper, lower and bite files (one or two bite scans) onto the window together, or click **Import scans** and choose them together. They go straight into the slots; a second file with *bite* in its name goes into **Second bite scan**.
2. Check the **Upper**, **Lower**, **Bite scan** and **Second bite scan** menus. Each lists every scan in the design and *None*.
3. If upper and lower came out the wrong way round, click the swap button (⇅) between them.
4. Read the line under the slots: ✓ *Bite as your scanner recorded it*, *Upper and lower are in occlusion*, or a warning (chapter 14).

The upper and lower scans turn grey (#919191) as they come in, and the models made from them are grey too; the bite scans keep the import blue. One arch is enough for a single model: leave the other slot on *None*. **New patient** starts a new design with Dental Studio open, so the next patient's scans cannot mix with these.

## Side by side, or in occlusion

Once the bite is found, the scans are laid out **side by side** on the plate - the upper on the left, the lower on the right, each teeth up with its front towards you - and the bite scan is hidden. **In occlusion** puts the pair in the bite to check it; **Side by side** puts them back for trimming. Neither changes the scans.




**In occlusion**: the two scans in the locked bite, to check it from the front and sides.

## How Pare 3D tells the upper from the lower

When you have not chosen the slots yourself, each scan is scored on three cues, strongest first: **which way the teeth point** (scanners place the upper above the lower, so its teeth point down - read on the file's vertical, whether the scanner exported it Z-up or Y-up), **a scanned palate**, and **the width across the canines**. For a pair in occlusion the choice is then checked by the overjet: the upper incisors must stand in front of the lower. Your own choice in the menus always wins.

## Full arch or quadrant

A full arch is a U: its middle is the tongue or palate, with no tooth there. A quadrant or sextant has its teeth at its middle. Pare 3D counts an arch as a quadrant only when at least 2% of its tooth surface lies within 3 mm of the middle of the teeth - a single scrap (a tag of tongue, or what a trim left) is not enough. Before build 75 one point was; on some scans that made every base come out as a quadrant's rectangular block.

## The cleaning report

Each arch is cleaned the moment it is chosen. The line under it reads, for example, *Holes filled: 2 · loose pieces removed: 14 · 18.5 mm of scan below the occlusal plane*.

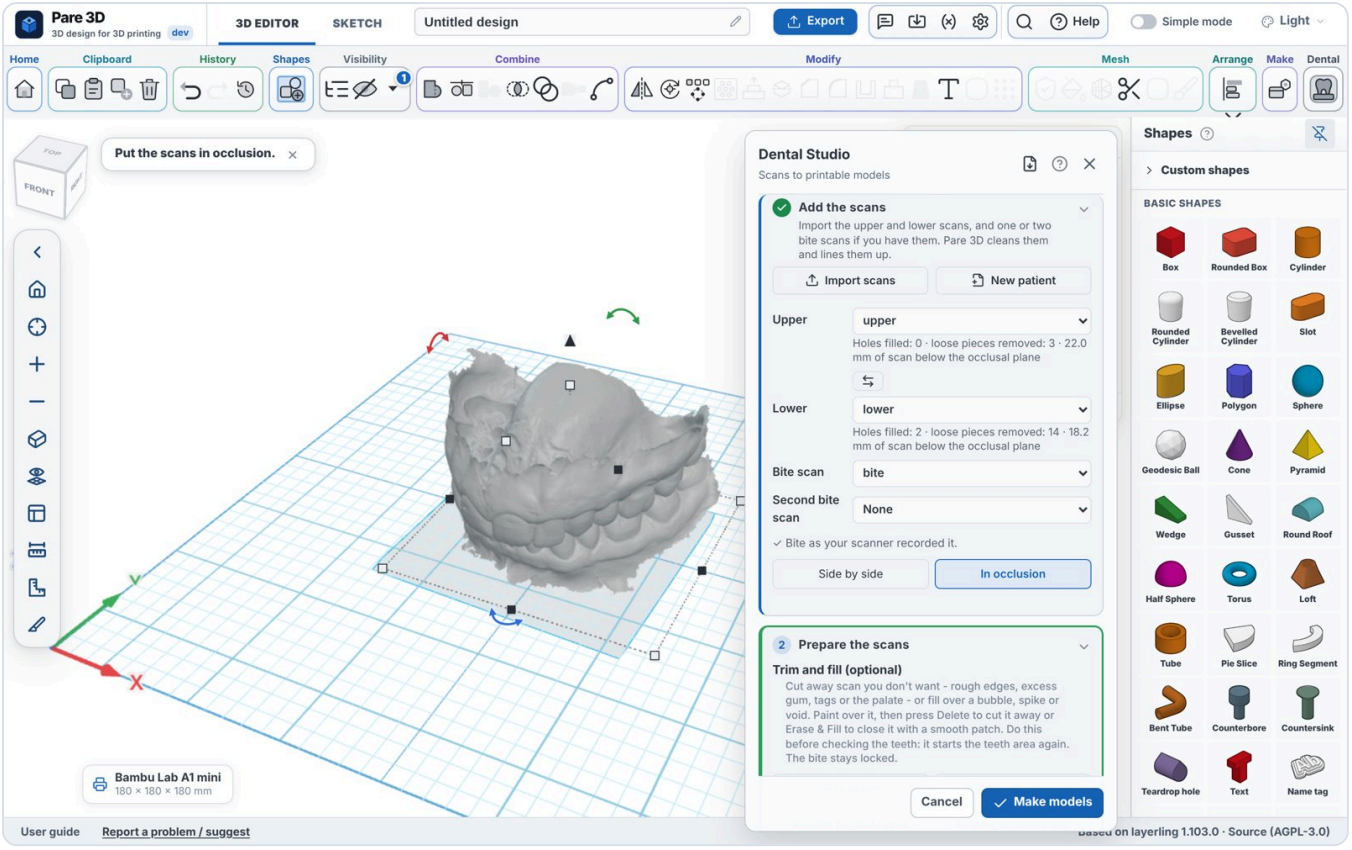
| Item | What was done |
|---|---|
| Holes filled | Every opening in the surface except the scan's outer edge (which becomes the margin side) is closed with new triangles - typically gaps between teeth or in the palate. |
| Loose pieces removed | Islands smaller than 5% of the scan - cheek, tongue, cotton roll and stray captures - are deleted. |

| | |
|---|---|
| Broken triangles | Edges shared by more than two triangles (non-manifold geometry) are removed so the model can be closed. |
| mm below the occlusal plane | How deep the scan reaches below the occlusal plane. This limits how much gum can be kept and how tall a base that *follows the scan* can be. |

Each arch is also set in a standard frame: teeth up, occlusal plane level, midline front to back, so the models come out level on the build plate. **Restore original upper** and **Restore original lower** (step 2) put a scan back exactly as it was imported, still in the locked bite.

# 5 The bite: one bite scan, two, or none

Scanners export the upper and lower arches already in occlusion, in the jaw relationship their own software set from the buccal bite records. Pare 3D keeps that bite exactly, as Medit's and 3Shape's model builders do, and uses the bite scans to check it.




The upper and lower arch scans in the scanner's bite (the bite scans themselves are hidden).

## Why not fit it again

Up to build 77, Pare 3D fitted both arches to the bite scan again. On our three test sets that moved the lower arch by 0.21 to 0.66 mm from where the scanner had put it - more than the published error of scanners' own bite records (about 50-135 µm on bilateral records in vitro [7-9]) - with no evidence that our placement was the better one. Since build 78 the scanner's bite is the default; your scanner's software is the reference, as it is for every other model builder.

## What the panel says

- ✓ *Bite as your scanner recorded it.* - the bite is kept, and every bite scan agrees with it. No figure is shown: a buccal bite scan carries scanning and stitching error of its own, typically 0.1-0.3 mm, so a small residual says nothing about the bite.
- Only when a bite scan clearly disagrees - more than **0.5 mm** RMS - does the line turn amber and offer **Line up on the bite scan** (**Line up on both bite scans** with two). Look at the occlusion from the front and both sides first.

## Two bite scans: one each side

Many clinicians record the bite on the right and on the left. Put the second in **Second bite scan**. The literature supports it: in a 2026 in-vitro study with three scanners, bilateral buccal registration was

significantly more accurate across the whole arch than a unilateral or an anterior record, whose accuracy fell with distance from where it was taken [16]; a right-only record was best on the right and worst elsewhere, and only the bilateral record was best for the full arch [17]; and bilateral bite scans beat one complete-arch bite scan (mean deviation 53 µm against 265 µm) [8].

- **Each bite scan is checked on its own** against the scanner's bite, so one side that disagrees cannot hide behind the other.
- **Lining up uses both together**, never one side alone. On one patient's scans (3Shape, a bite on each side), fitting to the right bite alone moved each arch about 0.3 mm; to the left bite alone, about 0.25 mm the other way; to both together, about 0.07 mm. A one-sided fit tilts the occlusion.

### Line up on the bite scans (when you ask)

Each arch is fitted to the bite scans separately, by **point-to-plane iterative closest point** (ICP) registration [18, 19], starting from where the scanner left it: about 8,000 points sampled over the arch, each paired with the nearest bite-scan point only if both surfaces face the same way (normals within about 53°, so an upper cusp never pairs with the lower cusp it touches), the rigid movement that best closes the distances solved and applied, up to 50 times, with the pairing distance narrowing from 2 to 1 to 0.5 mm. The fit is used only when both arches reach below **0.35 mm** RMS over enough shared surface. It is then locked into the scans.

### Without a bite scan

- If the two arches touch where the scanner put them, Pare 3D treats them as in occlusion (*Upper and lower are in occlusion*).
- If they do not touch, the models are still made, but without an articulator and with bases not matched to a bite: *The arches do not touch: add the bite scan to put them in occlusion*.
- If the two scans clearly do not share the scanner's coordinates (one was moved or turned in another program), each is made on its own, side by side.

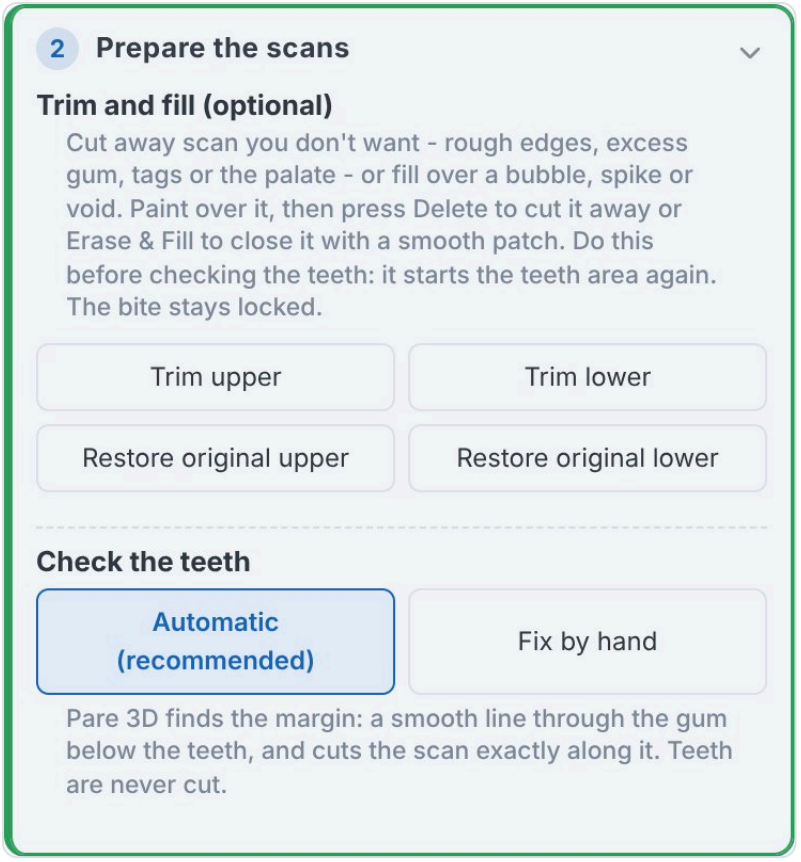
> **Checking the bite yourself**
> Turn the view with the right mouse button (or two fingers on a trackpad) and look from the front and both sides before pressing Make models. The models are made in exactly the relation you see.

### Kept through every edit

Each scan's frame - where it stands in the scanner's coordinates - is kept with it through moves, turns, trims, sculpting and cutting the palate away, together with each scan's original file. The models and the articulator are always made in the bite as exported (or as fitted, if you asked). **Restore original upper** and **Restore original lower** put a scan back as it was imported, still in the bite.

# 6 Step 2 - Prepare the scans: trim and fill, then check the teeth

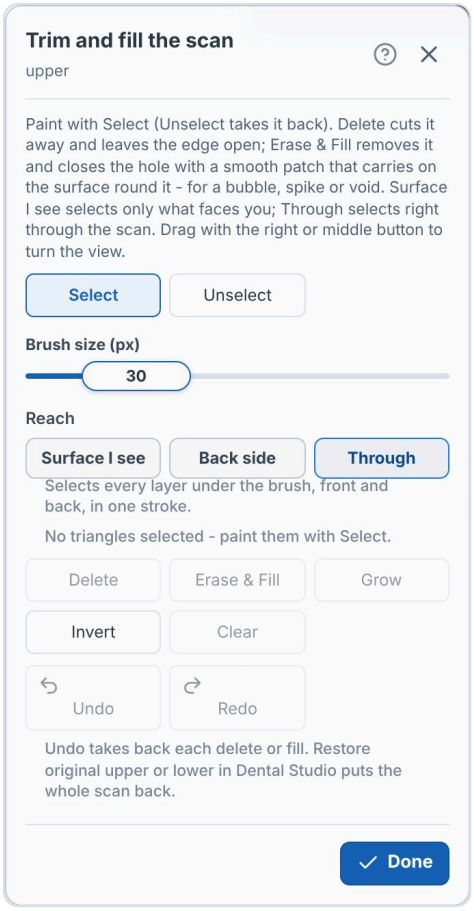
First cut away scan you do not want, or close over a defect; then check what Pare 3D keeps as teeth. In that order: trimming makes the scan anew, so the teeth area is found again after it.




Step 2: Trim and fill (optional) above, Check the teeth below.

## Trim and fill (optional)

**Trim upper** or **Trim lower** opens **Trim and fill the scan** on that arch, with only what this needs. The other scan steps aside while you trim, and the panel stands on the left of the work area, clear of Dental Studio.




Trim and fill: select, then delete or fill; undo and redo.

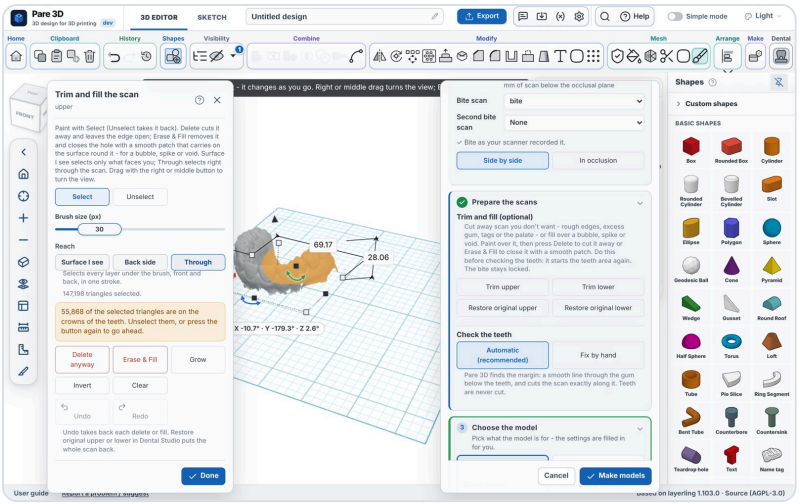
| Control | What it does |
|---|---|
| **Select** / **Unselect** | Paint over what to cut away; Unselect takes it back. |
| **Brush size** | In screen pixels. |
| **Surface I see** | Selects only what faces you - nothing hidden behind it. |
| **Back side** | The side facing away from you, at any depth. |
| **Through** (the default in Trim) | Every layer under the brush - the quick way to take a palate, with no hidden layer left behind. |
| **Delete** | Cuts the selection away and leaves the scan's edge open there. |
| **Erase & Fill** | Removes the selection and closes the hole with a smooth patch that carries on the surface round it (Liepa's hole filling, faired as a thin-plate surface[5,6], as Meshmixer does). For a bubble, spike or void. A hole that reaches the scan's own edge is left open. |
| **Undo** / **Redo** | Take back each delete or fill made since the panel opened, and do it again (`Ctrl` + `Z` / `Ctrl` + `Y` too). They never go back past the panel's opening. |
| **Grow**, **Invert**, **Clear** | Widen the selection by a ring of triangles, swap selected and not, or start again. |

Delete for rough or ragged edges, excess gum, a tag of cheek or tongue, or the palate when the model should be a horseshoe. Erase & Fill for a small defect on the gum or base: a bubble of saliva, a spike, a void where the scanner missed. Keep fills small - a few millimetres; over a large area the patch follows the curve of its rim and can bulge.

After each cut, anything a slicer would call non-manifold - an edge three triangles share, a corner where the scan only touches itself, a loose speck - is tidied away; the scan stays open.

**The crowns are guarded**

If what you selected takes in the crowns of the teeth, **Delete** and **Erase & Fill** say how many of the selected triangles lie on them and change nothing. Unselect them, or press the button again (**Delete anyway**) to go ahead. A crown is counted from the cusp tips 4 mm outwards and 6 mm down. Anatomical crowns are about 7.5 mm (molars) to 10-11 mm (incisors and canines) from cusp tip or incisal edge to the cemento-enamel junction, so this stays on the crown; on a short clinical crown it can reach the gingival margin. The gum, the palate and flaps of tissue do not count, nor fewer than 25 triangles at the edge of a stroke. A tall tag of tissue standing right behind the last molar can still be taken for a tooth.




A stroke drawn across the crowns: 34,568 of the 64,596 selected triangles lie on them, and nothing was cut.

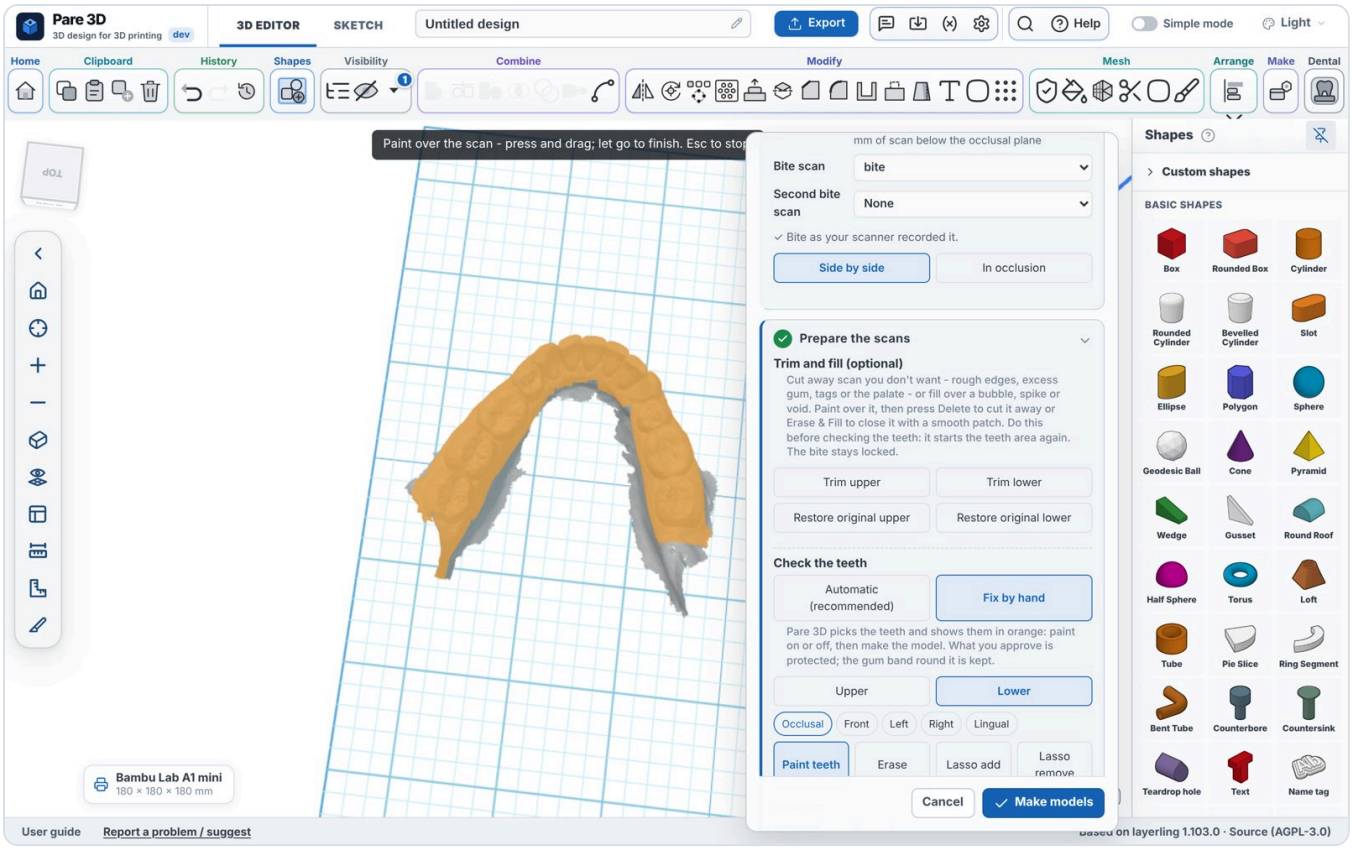
- **The bite stays locked** through every trim (chapter 5).
- **Undo** puts back what was cut or filled; **Restore original upper** / **lower** puts the whole scan back as it was imported.

## Check the teeth - Automatic (recommended)

Pare 3D finds the teeth, then lays a smooth line through the gum below them and cuts the scan exactly along it. Teeth are never cut. How far into the gum the line runs is set in step 4 (**Gum below the teeth**).

## Check the teeth - Fix by hand

Choose **Fix by hand** when the automatic choice missed a tooth surface (a deep gingival recession, a bridge pontic, an implant scan body) or took tissue for tooth (a frenum, a rugae fold). Pare 3D shows its choice as an **orange** tint on the scan.



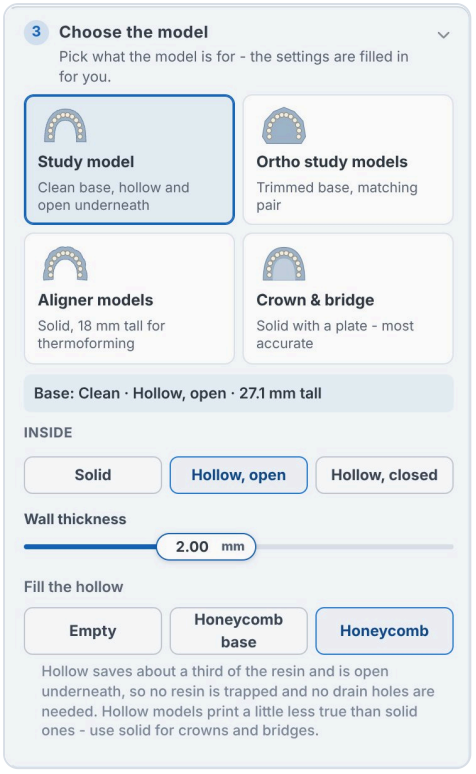

Fix by hand on the lower, from the occlusal side: orange is what the model keeps as teeth.

| Control | What it does |
|---|---|
| **Upper / Lower** | Which arch you are working on. |
| **Occlusal, Front, Left, Right, Lingual** | Turn the view to that side of the chosen arch. |
| **Paint teeth / Erase** | Brush the orange area on or off: drag over the surface. |
| **Lasso add / Lasso remove** | Draw a closed line round an area to add or remove it in one go. |
| **Expand into the gum** | The band of gum kept round the orange area (0.5-10 mm, default 3 mm). The scan is cut smoothly along its outer edge. |
| **Show the teeth area** | Hides the orange tint while you look at the scan; it is never shown while a scan is trimmed. |
| **Back to automatic** | Discards your changes on this arch. |
| **Sculpt this scan** | Opens Sculpt on this arch (chapter 12). |

The brushes change only the surface facing you. Turn the view between strokes with the right or middle mouse button; `Esc` puts the brush down. What you mark as tooth is **protected**: the cut never enters it.

# 7 Step 3 - Choose the model

Pick what the model is for; the base (step 4) and the inside are filled in for that purpose. You can change any of them afterwards.




Each choice shows its base from above; solid or hollow is chosen here too.

| Purpose | Base | Inside | Height |
|---|---|---|---|
| **Study model** | Clean: trimmed to a smooth outline a few mm round the teeth, on a low block | Hollow, open underneath | All anatomy kept, 3 mm base |
| **Ortho study models** | Orthodontic: trimmed the British/ABO way; a pair shares one back line | Solid | Base about a third of the model (6-15 mm) |
| **Aligner models** | Follows the scan | Solid | About 18 mm overall, for thermoforming |
| **Crown & bridge** | With a plate across the arch | Solid | 5 mm base |

Change the base in step 4 and the choice is marked *custom*; pick the model again to go back to its settings.

## Why these defaults

- **Study models** are hollow to save resin: about a third less. They are open underneath, so no resin is trapped and no drain holes are needed.
- **Aligner models** are solid and 15-20 mm tall, as Formlabs recommends for thermoforming: solid models reduce breakage in post-processing, and the height makes the formed appliance easier to remove [1].
- **Crown & bridge** models are solid with a plate across the arch. Hollow models are less true than solid ones, and in hollow models a thicker shell with a cross-arch bar gave the highest accuracy and stability after articulator mounting [2, 3]; solid with a plate is the safest choice where margins and contacts matter.
- **Ortho study models** follow the conventional British/ABO trimming: back square to the midline, heels, buccal sides at 65° (upper) and 55° (lower), the upper's front pointed and the lower's rounded, so the pair stands on its backs in occlusion.

## Inside

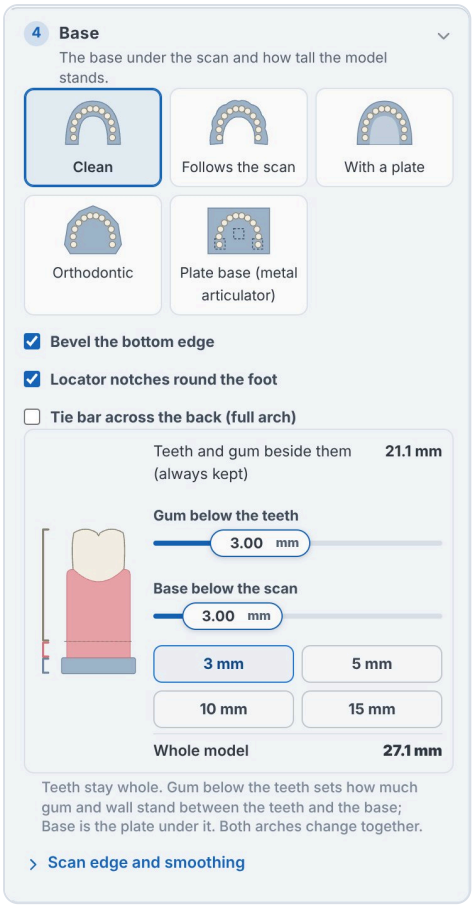
| Choice | What it is |
|---|---|
| **Solid** | Filled throughout. Prints truest. |
| **Hollow, open** | A shell open underneath: about a third less resin, nothing trapped, no drain holes needed. |

| | |
|---|---|
| **Hollow, closed** | A shell with a closed floor and vent holes up into the hollow, so resin drains and air gets in. Needed for a label underneath. |

- **Wall thickness** - 1-5 mm (default 2 mm). Formlabs advises at least 3 mm where a hollow model is thermoformed [1]; thicker shells were more accurate after mounting [2]. The wall is never thinner than set, also round the locator notches.
- **Fill the hollow** - *Empty* saves the most material; *Honeycomb base* stiffens the bottom only; *Honeycomb* carries the whole shell. All drain.

# 8 Step 4 - The base and the model's height

Always open: the base's shape and details, and how tall the model stands.

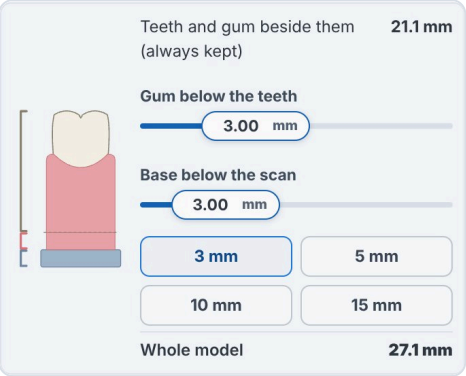



Base styles, details and the height.

| Style | Shape | Best for |
|---|---|---|
| **Clean** | The scan trimmed to a smooth outline a few mm round the teeth, standing on a low block | Study and patient-education models |
| **Follows the scan** | The base wall follows the scan's own edge straight down | Aligners and retainers |
| **With a plate** | A plate across the whole arch, palate or tongue space included | Crown and bridge, mounting - stiffest |
| **Orthodontic** | Trimmed art base (chapter 7) | Ortho records |
| **Plate base (metal articulator)** | A solid base with pockets underneath for a magnetic articulator plate | Mounting on a physical articulator; always solid |

- **Bevel the bottom edge** - a 0.8 mm bevel round the foot, so a scraper gets under it and the model comes off the build plate without chipping.
- **Locator notches round the foot** - half-round notches front and sides, so a lab can seat the model the same way each time. A hollow model keeps its full wall round them.
- **Tie bar across the back (full arch)** - joins the two back ends at the floor so a full arch cannot spread or warp while it prints. Cut it off afterwards.

## How tall the model is




A section through a molar in its gum on the base, each part bracketed to scale.

- **Teeth and gum beside them** - always kept: the crowns and the gum right beside them, down to where the deepest tooth reaches.
- **Gum below the teeth** - how much more gum and wall stand between the teeth and the base.
- **Base below the scan** - the block or plate under it, with 3, 5, 10 and 15 mm one click away.
- **Whole model** - the three together, from the floor to the tallest tooth. Both arches change together, and a pair stands equally tall.

The height is measured from the teeth, not from the tallest scrap of scan: on the author's set it reads 26.7 mm and the models came out 25.0 and 26.0 mm tall (before build 75 it read 34 mm).

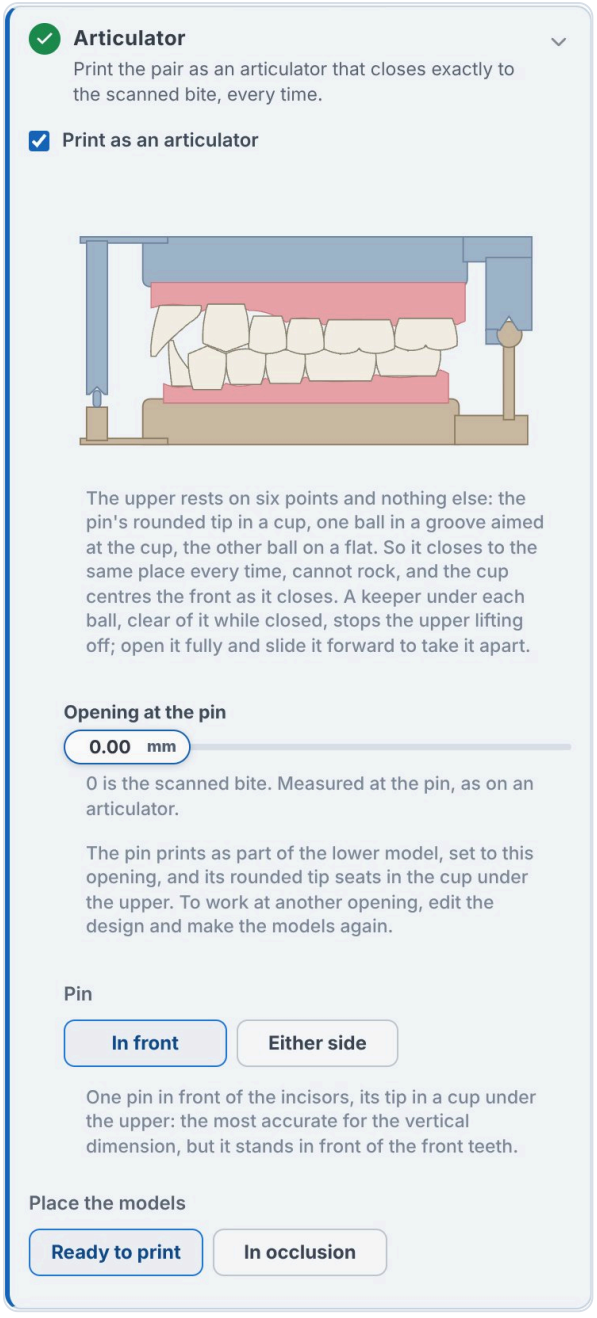
> **Wall too tall?**
> Lower **Gum below the teeth** first, then the base.

## Scan edge and smoothing

- **Follow the gum line (palate and floor of the mouth go)** - a band of gum all round the teeth and nothing more, so the model is a horseshoe; its depth is then the *Gum below the teeth* slider.
- **Smooth the scan's edge** - *Off*, *Low* (default) or *High*: how much the ragged scan border is smoothed where it meets the base. The teeth are never smoothed.

# 9 Step 5 - The articulator

For a pair only. Tick **Print as an articulator** and the two bases print as an articulator built so it cannot change the scanned bite. Choose where the pin stands and the opening; nothing else to set.




The articulator card: the side view, the opening at the pin, where the pin stands, and how the models are placed.

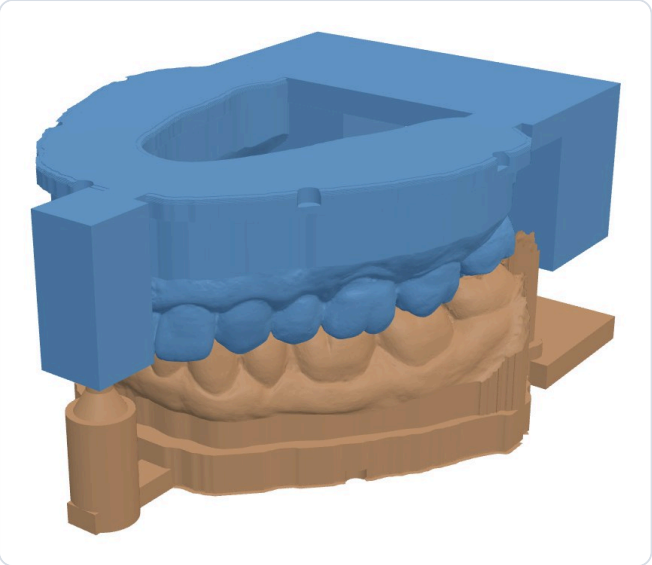
## How it holds the bite

The upper rests on **six points of contact and nothing else** - a kinematic coupling, the principle precision instruments use to come back to exactly the same position every time [10]. A rigid body has six degrees of freedom; six point contacts fix all six, no more, so nothing is held twice and nothing can bind or rock.

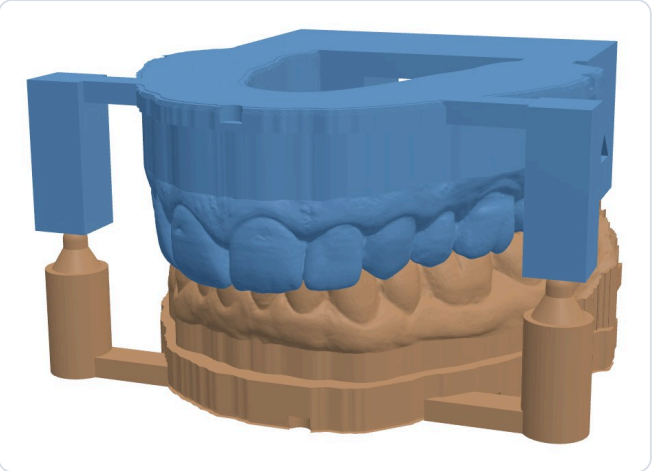
**Pin In front** (Kelvin's clamp):

- **The pin's rounded tip in a cup** - a 45° cone under the upper, as on a lab articulator: three points. As the models close, the cup centres the front of the upper on the pin.
- **A V-groove aimed at the cup** over one ball condyle: two points. Its ball can slide towards the pin, so the distance from condyle to pin never has to match exactly, whatever the printer shrinks.
- **A flat** over the other ball: one point. It stops the upper rolling and lets that ball sit wherever the print puts it.

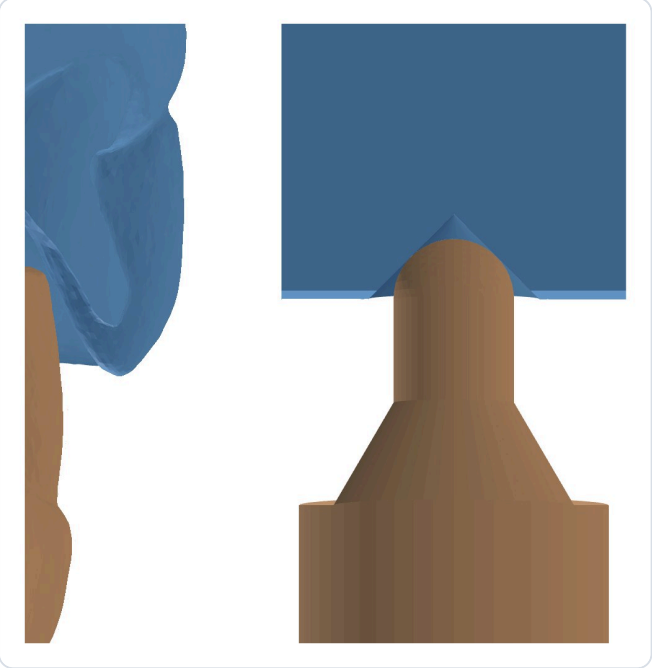
**Pin Either side**: a true hinge on the condyles - three faces over one ball, a groove along the axis over the other (five points) - and a post each side in the premolar region whose rounded tip rests on a flat pad under the upper. The posts only stop the hinge closing: the upper rests on whichever is a hair taller, and cannot rock.



**In front**: closed at 0, the scanned bite. The pin and its cup stand in front of the incisors.



**Either side**: a post each side in the premolar region; the front teeth are clear.



A section through the pin: its rounded tip seated in the cone-shaped cup under the upper.



The upper's condyle seats from below (pin in front): the flat (left) and the groove aimed at the pin (right), with the keepers' lips and channels.

## Which to choose

| | In front | Either side |
|---|---|---|
| Front teeth | The pin and its cup stand in front of the incisors | Clear to see, photograph, and seat an index on |
| Vertical accuracy | Highest: the pin is furthest from the hinge | The posts stand about halfway to the incisors, so a height error in a printed post shows about twice as large at the incisal edges |

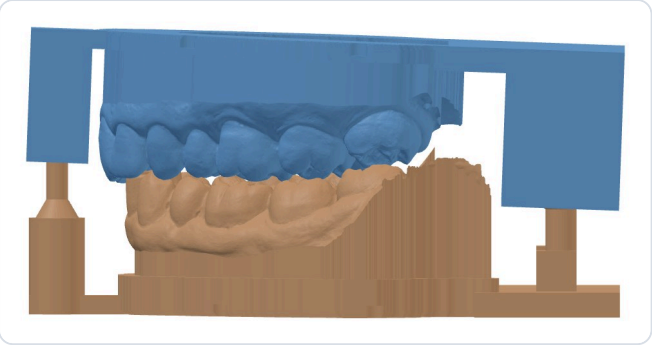
| | | |
|---|---|---|
| Width on the plate | Narrowest | About 8-10 mm wider each side |

**Never on a tooth**

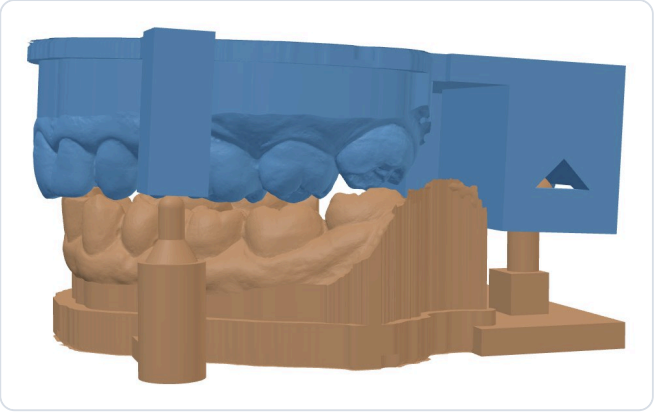
The pin and posts are placed from the furthest-forward (or furthest-out) point of either model across their band - teeth and bases included - and every part of them stands at least 2 mm clear of it, so proclined incisors or a wide arch never meet them. (In build 78 the pin was placed from the base, and on a patient with proclined incisors its pad met a front tooth.)

## The incisal pin and the opening

The pin (or each post) prints as part of the lower model: nothing to lose or fit. **Opening at the pin** sets the opening it is made at, 0 (the scanned bite) to 10 mm in half millimetres, read at the incisal pin's place in front of the incisors (with posts either side too, so the figure means the same). As on an articulator, the bite opens by **turning about the condyles**: the front teeth part by a little less than the pin and the back teeth by less again; after making, the notice says exactly how far, for example *the front teeth 2.7 mm and the back teeth 1.2 mm* at 3 mm. To work at another opening, use **Edit design** and make the models again (chapter 11).



Made at 3 mm, pin in front: the upper turned open about the condyles.



Made at 4 mm with a post either side.

**Above 6 mm**

The condyles stand on an average hinge axis, not the patient's. In a pilot study of 14 patients, average-value virtual mounting matched the patient's posterior opening at 3 and 6 mm increases at the incisors but not at 9 mm [11]. Confirm the vertical dimension clinically.

## No bands or clips

A **keeper** under each ball - a lip below its equator, 0.3 mm clear of it - never touches while the models are closed, so it cannot change the bite; it only stops the upper lifting off when the models are picked up or turned over. Held closed, the models sit exactly; pushed hard sideways the upper can ride up a seat by no more than that 0.3 mm before the keeper stops it, and settles back when let go. **To take it apart**, open it fully and slide the upper forward off the balls; to put it together, hold the upper open, slide it back onto the balls and close it. It opens to 90°.

**A quadrant against a full arch**

The pair is levelled and laid out by the full arch and articulated as a full pair: a condyle behind each side of the full arch and the pin in front of its central incisors (or the posts either side). The quadrant's base grows into a horseshoe 8 mm wide that follows the full arch's base, with the tongue space left open, and carries both condyles and the pin, so the models cannot rock or flex on the side with no opposing teeth.

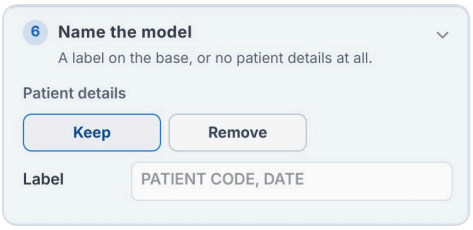
## What it does not do

It is a hinge: no protrusive or lateral excursions, no condylar guidance and no facebow transfer. Static virtual articulation - the scanner's bite record, which this articulator reproduces - describes the closed position only; evaluation methods vary and there is no agreed accuracy threshold [12]. For gnathological work - a full-mouth rehabilitation, say - mount the models on an adjustable articulator with a facebow (the **Plate base (metal articulator)** in step 4 makes a base for a metal articulator's magnetic plate).

After making, the notice says how far the articulator opens, e.g. *The articulator opens to 90° or more*; under 30° before the models meet, Pare 3D warns you to check it.

# 10 Step 6 - Name the model

A label on the base, or no patient details at all.




Patient details and the label.

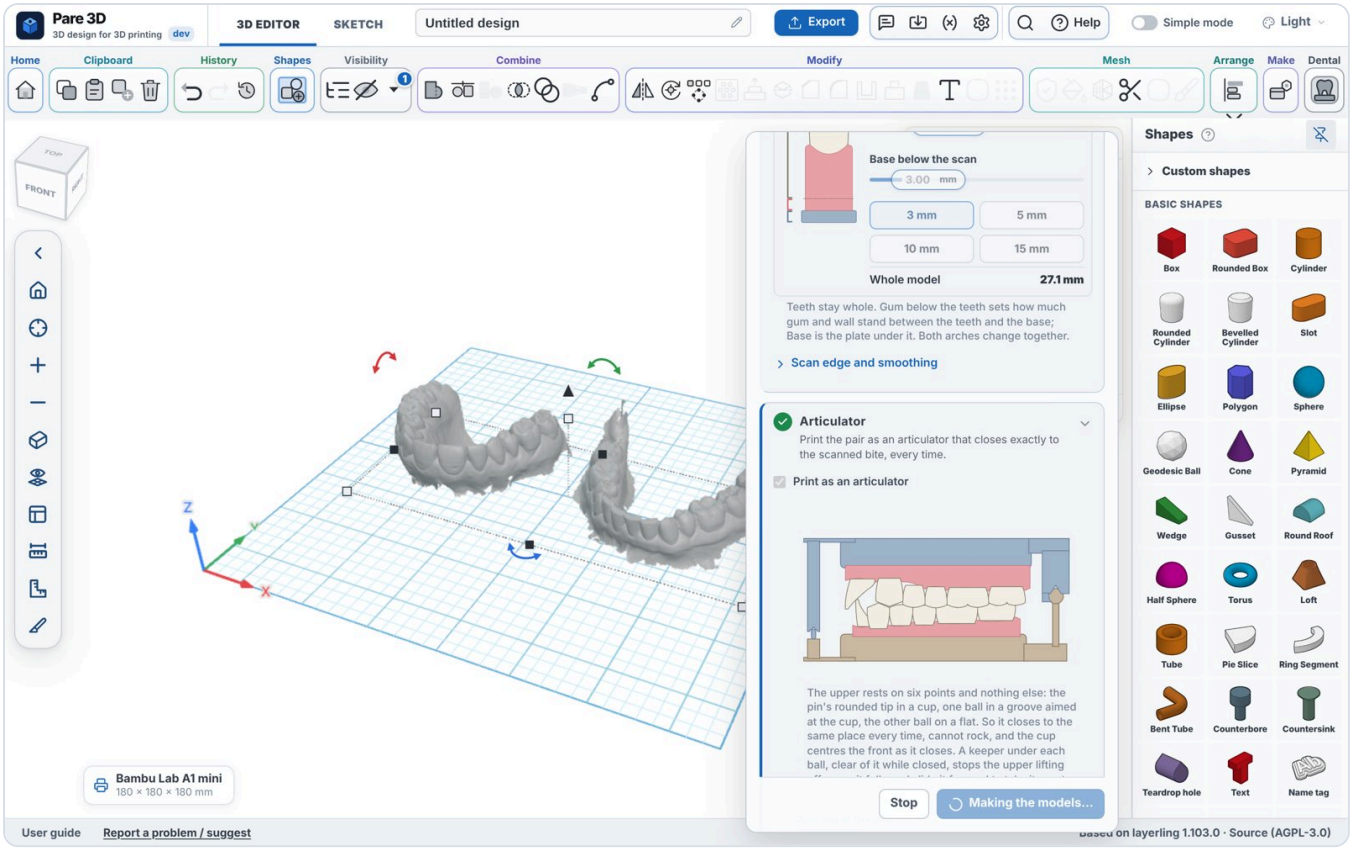
- **Patient details: Keep** names the models after the scan files (often the patient's name) and engraves the label.
- **Patient details: Remove** names them *Upper model* and *Lower model*, renames the bite scan and engraves no label, so the models and every file you export carry nothing that identifies the patient. Pare 3D remembers the choice.
- **Label** - a patient code or a date, engraved on the front of the orthodontic and articulator bases (with U or L in front for a pair) or under the floor on the others. A hollow model open underneath has nowhere to carry it, so with a label its floor is closed, with vent holes.

**One patient per design**

Keep each patient's scans in a design of their own: **New patient** in step 1 starts one with Dental Studio open. Give the design a case number rather than a name.

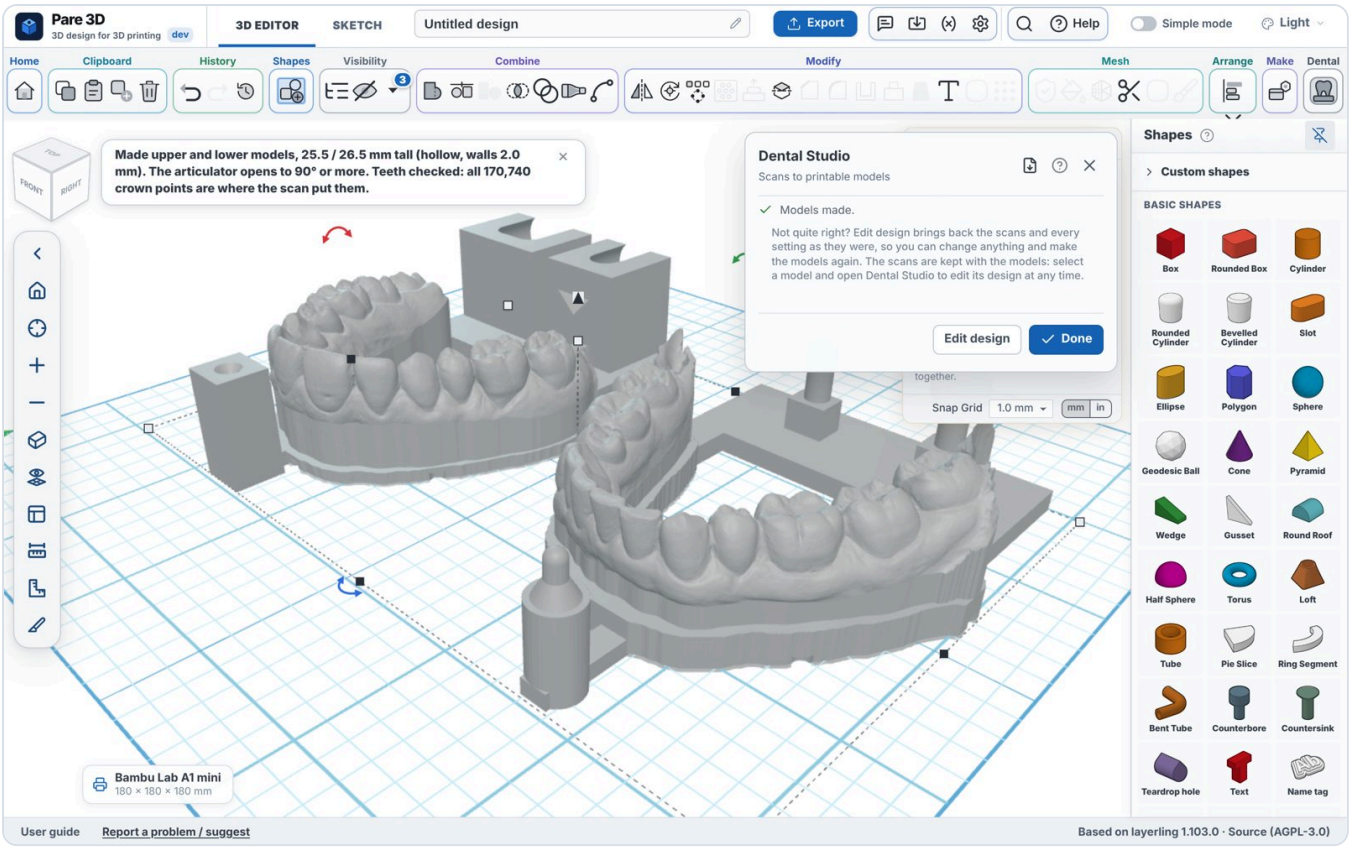
# 11 Making the models, and editing the design

Press **Make models** (or **Make model** for one arch). The button stays at the bottom of the panel, always on screen.

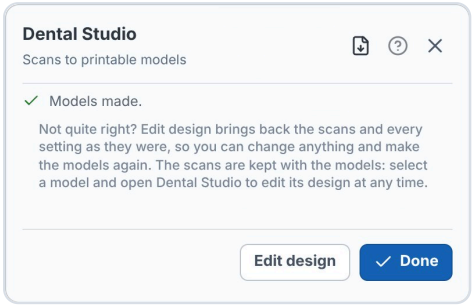



While the models are made you can still turn the view. **Stop** (the Cancel button while making) stops at once and leaves everything as it was.

The two arches are made at the same time in two background workers. Then the bases are matched, the articulator parts added, and the models placed. Before a model is handed over it is checked again: every crown point of the scan must be on the model where the scan put it.




The finished pair, ready to print: the condyles behind and the incisal pin in front.




After making, the panel stays open on the models.

## Not quite right? Edit design

- **The scans are kept.** The models are new parts; the scans and bite scans they were made from stay in the design, hidden, with every setting.
- **Edit design** brings them all back as they were when the models were made. Change anything - trim, the base, the opening, where the pin stands - and **Update models** makes the models again in place of the old ones.
- Close the panel instead and the models come back as they were.
- **Done** closes the panel. Later, select a model and open Dental Studio: its design opens again the same way.
- **Undo** (`Ctrl` / `Cmd` + `Z`) still takes back a make in one step.

The size of the selection is shown in the side panel; typing a size there scales the models - do not, if they must stay true to the patient.

# 12 Editing a scan by hand: Sculpt

Sculpt reshapes areas of a scan before the models are made - a spike, a fold of cheek, a rough area. To cut scan away, Trim (chapter 6) is simpler.

Open it from **Sculpt this scan** in Fix by hand, the brush button in the **Mesh** toolbar group, or the right-click menu. The panel opens at once and says *Getting the mesh ready to sculpt…* while a large scan is prepared (about a second for a full arch).

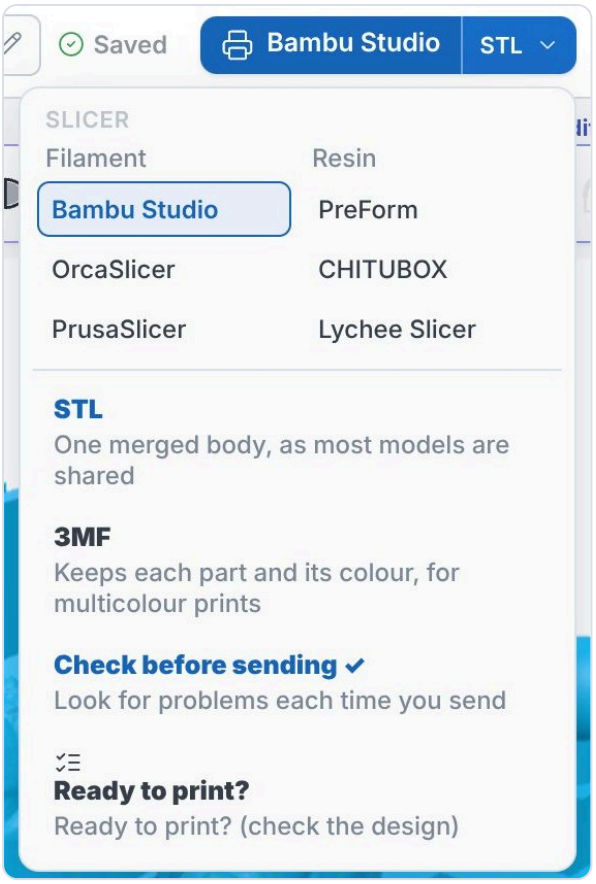
| Brush | Effect |
|---|---|
| Draw, Inflate, Deflate | Raise the surface, or push it out or in along its normals. |
| Smooth | Evens out bumps and spikes up to the brush's size; at full strength removes fissure detail. |
| Flatten, Pinch, Crease | Pull the surface onto a plane, together, or into a sharp line. |
| Move | Drag a patch of surface with the brush. |
| Select / Unselect | Paint triangles to delete, erase and fill, smooth or flatten as a group. |

- **Refine** adds triangles where the brush works, so fine detail can be shaped on a coarse part; **Symmetry** mirrors strokes left-right or front-back.
- Strokes show at once on the full-detail scan. **Undo** and **Redo** (or `Ctrl` / `Cmd` + `Z` and `Y`) step through strokes one at a time, then the deletes and fills made since the panel opened; the scan is rebuilt once when you press **Done**.
- Sculpting one of the Studio's scans hides the other until Sculpt closes; the bite stays locked.

> **Never sculpt the teeth you need to be true**
> Sculpt is for artefacts and soft tissue. Smoothing or flattening a tooth changes the anatomy the model reproduces.

# 13 Printing

Send the models straight to your slicer from the title bar.




The send menu: slicer and file type.

1. Open the small menu beside the Send button and choose the slicer: **Bambu Studio, OrcaSlicer, PrusaSlicer** (filament) or **PreForm, CHITUBOX, Lychee Slicer** (resin). A slicer not found on this computer is marked *not found*.
2. Choose **STL** (one merged body) or **3MF** (each part separately; the resin slicers read 3MF too).
3. Click **Send to ...**. The app saves a copy in *Documents › Pare 3D Exports* and opens it in the slicer.

With **Check before sending** on, *Ready to print?* first looks for floating parts, thin walls, large overhangs and parts past the build plate.

In a web browser the file is downloaded instead; open it in the slicer yourself.

## Printing tips

| | Resin (SLA / DLP / LCD) | Filament (FDM) |
|---|---|---|
| Orientation | Flat on the plate (Ready to print) for hollow open models; resin drains from underneath. Solid models can be printed flat with no supports. | Flat, teeth up. |
| Hollow | Use hollow open, or hollow closed with vents; never a closed shell without vents. | Solid or a honeycomb is usually faster than shells. |
| Articulator | Print flat, as laid out. Keep supports off the contact surfaces - the seats, the cup or pads under the upper, the top half of each ball and the tips of the pin or posts: printed flat these face away from the plate and need none, and a support mark there changes where the models close. Use your printer's finest validated layer height: the seats' accuracy comes down to round balls and flat faces. | Print flat, as laid out, with a fine layer height and no supports on the balls or seats; check the balls are round and the cup or pads smooth before relying on the bite. |
| Removal | Keep Bevel the bottom edge on. | - |

# 14 Messages and what to do

| Message | Meaning and what to do |
|---|---|
| The bite scan did not line up with the arches | One arch fitted the bite worse than 0.35 mm RMS, or they share too little surface. The scanner's positions are used. Check the bite from the front; re-export the bite scan from the same case if it is wrong. |
| The two scans are not where the scanner put them... | The arches do not share coordinates (one was moved or turned). Each is made on its own, without bite or hinge. Re-import the original exports. |
| The arches do not touch: add the bite scan... | No bite scan and the arches are apart. Add the bite scan in step 1; without it the models are made without an articulator. |
| One scan must be the upper and the other the lower | Both slots name the same arch. Use the swap button or the menus. |
| Part of the scan's ragged edge could not stand on the base... | The scan was cut where its edge is complete. Check that nothing you need was lost; rescan deeper if it was. |
| Some of the tooth surface could not be kept where the scan's edge folds under the teeth | An undercut or a fold at the edge. Check the model; trim the fold (step 2, Trim) and make it again. |
| There was no room for vent holes | A hollow closed model too small for vents. Use hollow open or solid. |
| The label was left off... | Labels need the orthodontic base, or a plate with a solid or closed inside, and room for letters at least 1.5 mm tall. |
| The bite scan does not match these arches well... | A bite scan disagrees with the scanner's bite by more than 0.5 mm. Check the occlusion from the front and sides; if it is wrong, use Line up on the bite scan (both bite scans, with two). |
| Stopped making the models. Nothing was changed. | You pressed Stop while making. Change what you want and make them again. |
| Teeth checked: ... % of ... crown points are where the scan put them | A few crown points were on specks of scan too small to print at a torn edge, or moved. The notice says how many; check that area of the model before you rely on it. |
| Teeth of one arch meet the other's gum in the bite (... mm²) | For example lower incisors on the palate in a deep overbite. Left untouched, so the printed models show the contact. Nothing to do; note it for the treatment plan. |
| Soft tissue cut back clear of the other model: ... mm² | Retromolar or tuberosity tissue that would have met the other model's soft tissue was trimmed 0.5 mm, as a technician trims a cast's heels. |
| The articulator opens less than 30° before the models meet | Something blocks the opening (tall distal tissue, a tie bar). Check it from the side before printing. |
| Part of a model came out loose from its base (a piece of the hinge or the stop) | With the articulator on: a piece of the hinge or the stop is not joined. Do not print; report it with the scans (Report a problem in the footer). |

| | |
|---|---|
| Part of a model came out loose from it - a piece of scan standing free of the model | Without an articulator: a piece of scan too big to be a mere tag stands free. Trim the scan there (step 2) and make the model again. |
| A small tag of scan standing free beside a model was left out | A sliver of cheek or tongue at the end of an arch, walled down to the floor on its own (under 150 mm³ and a twentieth of the model), held no teeth and was left out. Nothing to do. |
| … of the selected triangles are on the crowns of the teeth | Trim stopped before cutting or filling over crowns. Unselect them, or press the button again to go ahead. |
| Erased … triangles and filled … hole(s); … opening(s) ran into the scan's edge and were left open | Erase & Fill closed what it could; a hole that reaches the scan's own edge has nothing to close against. Use Delete there instead. |
| A large mesh - It has … triangles; cutting, grouping and merging handle … | The scan is over the triangle limit for cutting and grouping. Raise it (chapter 3); Dental Studio works either way. |
| The scans changed while the models were being made | A scan was edited during making. Make the models again. |

# 15 Quick reference

### The fastest route

1. Home › Dental models from scans
2. Drag the scan files onto the window (one or two bite scans)
3. Check: ✓ *Bite as your scanner recorded it*
4. Trim or fill if needed (step 2)
5. Pick the purpose
6. Tick Print as an articulator (pairs)
7. Make models
8. Send to your slicer

### Defaults by purpose

| | Base | Inside |
|---|---|---|
| Study | Clean, 3 mm | Hollow, open |
| Ortho | Orthodontic | Solid |
| Aligner | Follows scan, ~18 mm | Solid |
| C&B | Plate, 5 mm | Solid |

### The six steps

1 Add the scans · 2 Prepare (trim and fill, then teeth) · 3 Choose the model (and inside) · 4 Base and height · 5 Articulator (pairs) · 6 Name the model. Click a title to fold it. After making: Edit design or Done.

### Bite

Kept as the scanner exported it; each bite scan checked on its own, amber only above 0.5 mm. On request: point-to-plane ICP per arch on both bite scans together, normals must agree, 2 → 1 → 0.5 mm, up to 50 rounds, accepted below 0.35 mm RMS, then locked in.

### Articulator

Kinematic, six contacts. In front: pin tip in a cup, a groove aimed at it, a flat. Either side: hinge (three faces, axial groove) and two posts on pads. Pin part of the lower, 0-10 mm in 0.5 mm steps (warning above 6 mm), at least 2 mm clear of every tooth. Keepers, no bands; opens 90°.

### Keys

`Ctrl` + `K` search · `Ctrl` + `Z` undo (in Trim and Sculpt: back to the panel's opening) · `Esc` put a brush down · right or middle drag: turn the view · wheel: zoom.

# 16 Accuracy, and checking a printed articulator

## Accuracy

The Model Builder does not change the scanned tooth surfaces. Bases, hinges, pins and plates are added around the scan, and the teeth are carried through as the scanner recorded them. The accuracy of a finished model therefore depends mainly on:

- **The intraoral scan.** Trueness and precision are set by the scanner, the scanning technique and the clinical conditions. Refer to your scanner manufacturer's specifications.
- **The bite record.** By default, the Model Builder keeps the jaw relationship exactly as your scanning software exported it. Use **Line up on the bite scan** only when the arches arrive unaligned or the exported bite is clearly wrong; it fits each arch to the buccal bite scans together, so take one each side [16, 17]. Check the occlusion before printing. Published studies show digital bite records are generally more accurate than conventional silicone or wax records [9]. Accuracy falls with fewer teeth, quadrant records, large edentulous spans and scans taken away from the bite record site [13].
- **The printer, resin and post-processing.** Dimensional accuracy depends on the printer's calibration, layer height, resin, orientation, washing and curing. Follow the resin manufacturer's validated settings.
- **The articulator.** Hinge models reproduce the closed intercuspal position only. They do not record or reproduce jaw movements, condylar guidance or a facebow transfer.

Before you use a printed model for any restoration, appliance or diagnosis, check the result. Compare the on-screen occlusion with the patient's scans, confirm the printed teeth seat correctly against each other, and verify critical fits clinically. Pare 3D is a model-building tool, not a medical device. The clinician remains responsible for confirming that a model is fit for its intended use.

## What Pare 3D checks for you

- **The teeth.** Every crown point of each scan is checked against the finished model before it is handed over. On the author's scans and six cut-down versions of them, every one of 26,000 to 93,000 crown points per arch was within 0.002 mm of where the scan put it (the limit of the 32-bit numbers the file holds).
- **Soft tissue in the way.** Soft tissue behind the last teeth - the retromolar pad and the tuberosity - that would meet the other model's soft tissue in the bite is cut back 0.5 mm, as a technician trims a cast's heels; otherwise it, not the teeth, would hold the models open. Gum that the other arch's teeth meet, such as the palate under a deep overbite, is never trimmed: that contact tells you how far the bite can be opened. The orthodontic records preset keeps the scans exactly as captured, since any alteration of tooth or soft tissue anatomy invalidates board records [14].
- **Where the scans overlap.** Where the scanner's own bite has teeth overlapping slightly (registration error in the bite record), the printed teeth touch there first, as stone casts from the same records would. On a fixed hinge axis this cannot make the models rock; it can only stop them closing a fraction early.

## Checking a printed articulator

No manufacturer publishes how accurately its printed articulator closes, and there is no ISO standard for printed models or articulators. In the one study that measured articulated printed casts, casts on CAD-designed articulation pins were about 333 µm RMS out between the arches, against about 149 µm when articulated by hand [15]. To know for your own printer and resin, this is the protocol Pare 3D is being validated with (in vitro, on a typodont or a patient's casts with consent):

1. **Reference.** Scan a set of arches with a bilateral buccal bite as usual. Mark the canine cusp tips and the first molars' mesiobuccal cusp tips on the scans.

2. **Print.** Make the articulated pair, print it flat with your printer and resin's validated settings, wash and cure as directed, and measure within a day.
3. **Repeatability.** Open and close it 20 times; each time measure the gap at the pin (or a post) and at a molar with a dial indicator or feeler gauges. Target: standard deviation 25 µm or less.
4. **Trueness of the bite.** Scan the closed pair with a lab or intraoral scanner, register each printed arch to its source scan, and compare the distances between the marked points across the arches with those in the scanned bite. Target: 150 µm RMS or less (stretch: 100 µm).
5. **Contacts.** At the contacts the screen shows, 12 µm shimstock should hold; where it shows none, it should pull through.
6. **Pin steps.** With models made at 2, 4 and 6 mm, measure the opening at the incisal edges and the first molars. Target: within 0.15 mm of the figures in the notice.
7. **Durability.** Repeat step 3 after 500 open-close cycles, and after 7 and 14 days.

Publish the printer, resin, settings and every measurement with the results, so others can repeat them.

# 17 References

Every source the Model Builder has drawn on, from the first build to this one: printing and model accuracy (1-3), condylar guidance (4), mesh repair (5, 6), bite registration and virtual articulation (7-9, 11-13, 16, 17), the articulator's kinematic coupling (10) and printed articulation (15), records standards (14), registration methods (18, 19), smile design and injection moulding, for the wax-up module in development (20-30), and the geometry methods and libraries used (31-34).


1. Formlabs. Thermoformed clear aligners and retainers - application guide. *dental.formlabs.com/indications/thermoformed-clear-aligners-retainers/guide* (accessed 26 October 2026). Models 15-20 mm tall; solid models recommended; hollow walls at least 3 mm.
2. Wan Q, Riera C, Tran TT, Fehmer V, Sailer I, Lee JH. Printing accuracy and post-mounting dimensional stability of 3D-printed hollow dental models with varying shell thicknesses and cross-arch bars. *J Dent*. 2026;164:106234. doi:10.1016/j.jdent.2025.106234
3. Yurdakurban E, Topsakal KG, Duran GS, Görgülü S. Evaluating the accuracy between hollow and solid dental aligner models: a comparative study of printing technologies. *Aust Orthod J*. 2024;40(2):51-62. doi:10.2478/aoj-2024-0023
4. Kwon OK, Yang SW, Kim JH. Correlation between sagittal condylar guidance angles obtained using radiographic and protrusive occlusal record methods. *J Adv Prosthodont*. 2017;9(4):302-307. doi:10.4047/jap.2017.9.4.302 (protrusive records: 30.1° ± 6.7° right, 30.2° ± 6.3° left).
5. Liepa P. Filling holes in meshes. In: *Proceedings of the Eurographics/ACM SIGGRAPH Symposium on Geometry Processing*. Aachen: Eurographics Association; 2003:200-206. doi:10.2312/SGP/SGP03/200-206
6. Kobbelt L, Campagna S, Vorsatz J, Seidel HP. Interactive multi-resolution modeling on arbitrary meshes. In: *Proceedings of SIGGRAPH 98*. New York: ACM; 1998:105-114. doi:10.1145/280814.280831
7. Revilla-León M, Gómez-Polo M, Zeitler JM, Barmak AB, Kois JC, Pérez-Barquero JA. Does the available interocclusal space influence the accuracy of the maxillomandibular relationship captured with an intraoral scanner? *J Prosthet Dent*. 2024;132(2):435-440. doi:10.1016/j.prosdent.2022.09.004 (bilateral record, TRIOS 4: trueness 135 µm at maximum intercuspation, 71-73 µm with the pin opened 3-4°).
8. Lee JD, Luu D, Yoon TW, Lee SJ. Accuracy comparison of bilateral versus complete arch interocclusal registration scans for virtual articulation. *J Prosthet Dent*. 2024;132(5):1038-1043. doi:10.1016/j.prosdent.2023.01.028 (bilateral 53 ± 51 µm; complete arch 265 ± 241 µm).
9. Ries JM, Grünler C, Wichmann M, Matta RE. Three-dimensional analysis of the accuracy of conventional and completely digital interocclusal registration methods. *J Prosthet Dent*. 2022;128(5):994-1000. doi:10.1016/j.prosdent.2021.03.005 (intraoral scanner 0.05 ± 0.04 mm; silicone records 0.41-0.44 mm).
10. Slocum AH. Kinematic couplings: a review of design principles and applications. *Int J Mach Tools Manuf*. 2010;50(4):310-327.
11. Lin YC, Scialabba R, Lee JD, Sun J, Lee SJ. Assessment of occlusal vertical dimension change in mechanical and virtual articulation: a pilot study. *Dent J (Basel)*. 2022;10(11):212. doi:10.3390/dj10110212
12. Zheng Y, Zhang T, Liu Y, Pradíes G, Wang X, Ma D, Bai S. Evaluation methods and influencing factors on accuracy for static virtual articulation obtained by intraoral scanners: a critical review of the literature. *J Esthet Restor Dent*. 2025;37(2):378-399. doi:10.1111/jerd.13312
13. Revilla-León M, Alonso Pérez-Barquero J, Zubizarreta-Macho Á, Barmak AB, Att W, Kois JC. Influence of the number of teeth and location of the virtual occlusal record on the accuracy of the maxillo-mandibular relationship obtained by using an intraoral scanner. *J Prosthodont*. 2023;32(3):253-258. doi:10.1111/jopr.13526
14. American Board of Orthodontics. ABO digital model requirements. exam.americanboardortho.com (accessed 27 October 2026): "Alteration of tooth or soft tissue anatomy, in any form using any method, is considered record falsification."
15. Alhinti O, Kudara Y, Pagni S, El Rafie K, Vazouras K, Ferrari M, Chen YW. Accuracy of occlusal relationships in 3D-printed dental casts: a comparative in vitro study. *J Prosthodont*. 2025. doi:10.1111/jopr.70039 (hand articulation 149.4 µm RMS; CAD-generated articulation pins 332.8 µm).
16. Matta RE, Förtsch F, Pfeuffer S, Wichmann M, Ries J. Accuracy of three different strategies for virtual interocclusal registration: an in vitro study. *J Esthet Restor Dent*. 2026;38(11):2192-2198. doi:10.1111/jerd.70213 (bilateral buccal registration most accurate across the arch; unilateral and anterior lose precision with distance from the scanned site).
17. Mei J, Ma L, Chao J, Liu F, Shen J. Three-dimensional analysis of the outcome of different scanning strategies in virtual interocclusal registration. *J Adv Prosthodont*. 2022;14(6):369-378. doi:10.4047/jap.2022.14.6.369 (bilateral posterior record: smallest full-arch deviation).
18. Besl PJ, McKay ND. A method for registration of 3-D shapes. *IEEE Trans Pattern Anal Mach Intell*. 1992;14(2):239-256. doi:10.1109/34.121791

19. Chen Y, Medioni G. Object modelling by registration of multiple range images. *Image Vis Comput*. 1992;10(3):145-155. doi:10.1016/0262-8856(92)90066-C (point-to-plane registration).
20. Magne P, Gallucci GO, Belser UC. Anatomic crown width/length ratios of unworn and worn maxillary teeth in white subjects. *J Prosthet Dent*. 2003;89(5):453-461. doi:10.1016/S0022-3913(03)00125-2
21. Chu SJ, Tan JH, Stappert CF, Tarnow DP. Gingival zenith positions and levels of the maxillary anterior dentition. *J Esthet Restor Dent*. 2009;21(2):113-120. doi:10.1111/j.1708-8240.2009.00242.x
22. Coachman C, Calamita M. Digital smile design: a tool for treatment planning and communication in esthetic dentistry. *Quintessence Dent Technol*. 2012;35:103-111.
23. Coachman C, Calamita MA, Coachman FG, Coachman RG, Sesma N. Facially generated and cephalometric guided 3D digital design for complete mouth implant rehabilitation: a clinical report. *J Prosthet Dent*. 2017;117(5):577-586. doi:10.1016/j.prosdent.2016.09.005
24. Coachman C, Calamita MA, Sesma N. Dynamic documentation of the smile and the 2D/3D digital smile design process. *Int J Periodontics Restorative Dent*. 2017;37(2):183-193. doi:10.11607/prd.2911
25. Sarver DM. The importance of incisor positioning in the esthetic smile: the smile arc. *Am J Orthod Dentofacial Orthop*. 2001;120(2):98-111. doi:10.1067/mod.2001.114301
26. Magne P, Belser UC. Novel porcelain laminate preparation approach driven by a diagnostic mock-up. *J Esthet Restor Dent*. 2004;16(1):7-16. doi:10.1111/j.1708-8240.2004.tb00444.x
27. Gürel G. Porcelain laminate veneers: minimal tooth preparation by design. *Dent Clin North Am*. 2007;51(2):419-431. doi:10.1016/j.cden.2007.03.007
28. GC Australasia; Tay A. Injection moulding: case study and technique guide (G-ænial Universal Injectable, EXACLEAR). Banksmeadow (NSW): GC Australasia; n.d. www.gc.dental/australasia (technique-guide-injection-moulding-exaclear.pdf; accessed 28 October 2026).
29. GC Europe. GC Get Connected special edition: injection moulding technique. GC Europe N.V.; 2023. www.gc.dental/europe (GCGC Special Edition IMT; accessed 28 October 2026).
30. Spadoni D, Valeri C, Quinzi V, Schneider Moser U, Marzo G. Advancing orthodontic aesthetics: exploring the potential of injectable composite resin techniques for enhanced smile transformations. *Dent J (Basel)*. 2025;13(1):18. doi:10.3390/dj13010018
31. Lorensen WE, Cline HE. Marching cubes: a high resolution 3D surface construction algorithm. *SIGGRAPH Comput Graph*. 1987;21(4):163-169. doi:10.1145/37402.37422 (the hollow models' inner surface).
32. Möller T, Trumbore B. Fast, minimum storage ray-triangle intersection. *J Graph Tools*. 1997;2(1):21-28. doi:10.1080/10867651.1997.10487468 (picking points on a scan).
33. Kapoulkine A. meshoptimizer [software]. MIT licence. github.com/zeux/meshoptimizer (the tooth library's simplification; accessed 28 October 2026).
34. Teethy. Anterior tooth library. teethy.org (redistributed in Pare 3D with the owner's written permission; accessed 28 October 2026).

The registration, cleaning, margin and articulator methods described here are those implemented in Pare 3D build 80 (files *dentalArch*, *dentalModel* and *dentalArticulator* in the source code, AGPL-3.0). Pare 3D is not a medical device; check every model before clinical use.